

A New Method for the Evaluation of Small-Angle Scattering Data

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A new numerical method is presented for simultaneous smoothing, desmearing and Fourier transformation of X-ray and neutron small-angle scattering data. The method can only be applied to scattering curves from dilute particle systems, *i.e.* for scattering media whose distance distributions are zero beyond a certain value. The distance distribution of the scattering medium is approximated by a linear combination of about 20 to 30 cubic *B*-splines. These spline functions have a restricted extension in real space. Their coefficients are adjusted by a weighted least-squares operation so that the series, after being Fourier transformed and smeared according to the geometry and wavelength distribution, represents an optimum smoothed approximation of the experimental data. Tendencies towards oscillations in the least-squares operation are suppressed by a new stabilization routine. The method offers a new possibility for the estimation of the radius of gyration, which is generally superior to the Guinier approximation.

Introduction

The relation between a particle and its scattering data is shown schematically in Fig. 1. We discuss non-oriented particles. By means of scattering we can obtain information only about their distance distribution, $p(r)$. The physical process of scattering can be mathematically expressed by a Fourier transformation of $p(r)$ (transformation T_1). In the particular case of spherical, symmetric particles the radial electron-density profile is correlated in the same way with the scattering amplitude, the square of which is the measurable scattering intensity.

In practice, the scattering intensity cannot be measured exactly. A monochromatic primary beam with point collimation of very high intensity and an infinitely small detector would be necessary to obtain ideal data. The influence of polychromatic radiation is described by the linear transformation T_2 , while the transformations T_3 and T_4 represent the slit-length and slit-width effects. In practice, some of these effects can be neglected.

The measuring range is limited at small angles by the finite dimensions of the primary beam and the width of the detector window. At larger angles, the scattered intensity drops down to small values. This leads to small counting rates. Subtracting the background leads to a further increase of statistical errors. Altogether, a 'picture' of the investigated particle is obtained which is a multiple transform of the distance distribution. The restricted measuring range and statistical errors reduce the significant information.

A new method which allows accurate calculation of the distance distribution $p(r)$ from the measured scattering data is presented. No preliminary data preparation is necessary, except extrapolation to infinite dilution (elimination of interparticle scattering). The desmeared smoothed scattering curve is calculated as an intermediate result. The method needs rough *a priori*

information about the maximum distance in the particle. This is one reason why this method cannot be used for periodic scattering systems.

Mathematical description

The 'characteristic' of a particle $\gamma(r)$, the so-called correlation function, was introduced in the theory of small-angle scattering by Porod (1951). Multiplying by r^2 we obtain the distance distribution function $p(r)$

$$p(r) = \gamma(r)r^2 \quad (1)$$

(Porod, 1948; Guinier & Fournet, 1955, p. 12). The square of the radius of gyration is half the normalized second moment of this function (Guinier & Fournet, 1955, p. 28)

$$R_g^2 = \int_0^\infty p(r)r^2 dr / 2 \int_0^\infty p(r) dr. \quad (2)$$

The scattering intensity at zero angle is a constant, 4π , times the integral of $p(r)$:

$$I(0) = 4\pi \int_0^\infty p(r) dr. \quad (3)$$

These two formulae will be used later to find new approximations for R_g and $I(0)$. We shall now describe the various transformations in more detail.

T_1 : The relation between the scattering intensity and the characteristic of a particle is given by the Fourier transformation:

$$I(h) = 4\pi \int_0^\infty p(r) \frac{\sin(hr)}{hr} dr \quad (4)$$

where

$$h = (4\pi/\lambda_0) \sin \theta, \quad (5)$$

λ_0 is the wavelength of the monochromatic radiation and 2θ is the scattering angle.

T_2 : The experimentally observed scattering intensity

I_m , measured with polychromatic radiation, is connected with the theoretical scattering intensity I for monochromatic radiation of wavelength λ_0 as follows:

$$I_m(h) = \int_0^\infty W(\lambda') I(h/\lambda') d\lambda' \quad (6)$$

where $\lambda' = \lambda/\lambda_0$, $W(\lambda')$ is the wavelength distribution of the polychromatic radiation.

T_3, T_4 : The two geometric effects (T_3 and T_4) have been described in detail elsewhere (Kratky, Porod & Skala, 1960).

Summarizing all effects ($T_1 - T_4$), the following relation for the experimentally observed scattering intensity, $I_{\text{exp}}(h)$, is obtained:

$$I_{\text{exp}}(h) = \left[8\pi \int_0^\infty dr \int_0^\infty d\lambda' \int_0^\infty dt \int_{-\infty}^\infty dx \right. \\ \left. \times Q(x) P(t) W(\lambda') p(r) \sin(\beta)/\beta \right] + \varepsilon(h), \\ \beta = r[h - x]^2 + t^2]^{1/2}/\lambda' \quad (7)$$

where $P(t)$ is the weighting function for the slit-length effect and $Q(x)$ is the weighting function for the slit-width effect. $\varepsilon(h)$ is the statistical error corresponding to the pulse rate of the counter.

Method

The method for the solution of this integral equation presented here corresponds to the indirect Fourier-transformation method, which has been published

recently (Glatter, 1977). We start with the assumption that a value D can be estimated for which

$$p(r) = 0 \quad \text{for } r \geq D. \quad (8)$$

The function $p(r)$ thus differs from zero only in a certain limited region of real space, namely between 0 and D , where D corresponds to the maximum distance in the particle. The function $p(r)$ is approximated in this range by a linear combination of a finite number of functions $\varphi_v(r)$:

$$p_A(r) = \sum_{v=1}^N c_v \varphi_v(r) \quad (9)$$

where the $\varphi_v(r)$ are cubic B -splines (Greville, 1969; Schelten & Hossfeld, 1971). These functions are an appropriate system. They are defined as convolution products of a step function and represent curves with minimum second derivative. They overlap only with a few neighbours as they differ from zero only in the range of four knots. The expansion coefficients c_v are the unknowns.

The particular functions φ_v are now transformed according to the transformations in the correct sequence (T_1 to T_4) (Glatter & Zipper, 1975). The result after the transformation T_1 represents scattering intensities without collimation effects:

$$\psi_v(h) = T_1 \varphi_v(r). \quad (10)$$

Smeared intensities result after execution of all transformations

$$\chi_v(h) = T_4 T_3 T_2 T_1 \varphi_v(r) = T_4 T_3 T_2 \psi_v(h). \quad (11)$$

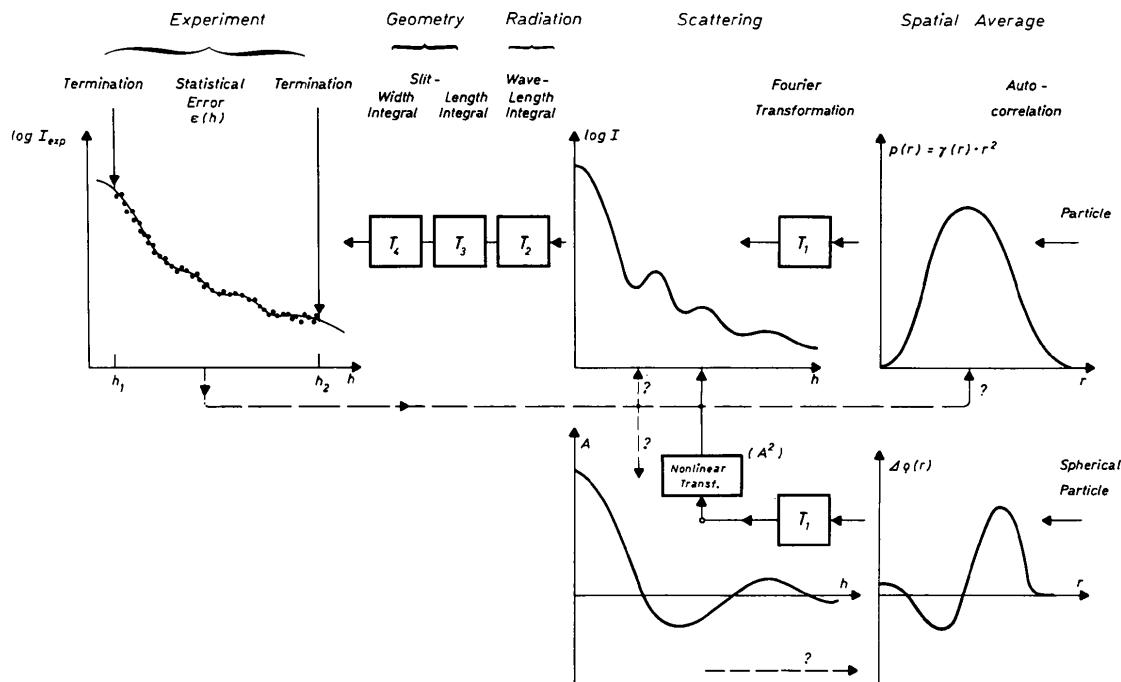


Fig. 1. Schematic representation of the correlation between a particle and its observable experimental scattering data.

The coefficients c_v are now determined by a weighted least-squares approximation of the experimental data $I_{\text{exp}}(h)$ by the sum $\sum c_v \chi_v(h)$. This condition requires that

$$L = \sum_{i=1}^M [I_{\text{exp}}(h_i) - \sum_{v=1}^N c_v \chi_v(h_i)]^2 / \sigma^2(h_i) = \min., \quad (12)$$

where M is the number of data points and $\sigma(h_i)^2$ is the variance of the i th point. The solution of the minimization problem in (12) leads to the normal equations

$$B_{\mu\nu} c_\nu = d_\mu \quad \mu, \nu = 1, 2, \dots, N \quad (13)$$

where

$$B_{\mu\nu} = \sum_{i=1}^M \chi_\mu(h_i) \chi_\nu(h_i) / \sigma^2(h_i)$$

and

$$d_\mu = \sum_{i=1}^M I_{\text{exp}}(h_i) \chi_\mu(h_i) / \sigma^2(h_i).$$

As N may not be chosen too small, so as to guarantee a sufficiently exact description of $p(r)$, and as the data points always have statistical errors, stability problems usually arise in the solution of (13). This difficulty can be overcome by the modified condition

$$N_{c'} = \sum_{v=1}^{N-1} (c_{v+1} - c_v)^2 \quad (L + \lambda N_{c'}) = \min. \quad (14)$$

This constraint changes (13) into

$$(B_{\mu\nu} + \lambda K_{\mu\nu}) c_\nu = d_\mu \quad (15)$$

where the matrix $K_{\mu\nu}$ is defined as

$$K = \begin{pmatrix} 1 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & -1 & \\ & & & \ddots & \\ & & & -1 & 2 & -1 \\ & & & & -1 & 1 \end{pmatrix}.$$

The stabilizing parameter, λ , indicates how much stabilization is taken into account.

Point of inflexion method for the determination of λ_{opt}

To obtain an optimum value of λ we start with solving (15) for a series of different values of λ . The resulting dependence of the mean deviation per point

$$\text{MDP} = \frac{1}{M} \sum_{i=1}^M [I_{\text{exp}}(h_i) - \sum_{v=1}^N c_v \chi_v(h_i)]^2 / \sigma^2(h_i) \quad (16)$$

and the influence of λ on $N_{c'}$ is shown in Fig. 2 for a particular example. The function $\log N_{c'}(\lambda)$ shows a point of inflexion in the region close to the beginning of a strong increase of the mean deviation (overstabilization). The value of λ which corresponds to this point of inflexion (absolute value of the first derivative has a minimum) is the optimum value λ_{opt} with sufficient stabilization and negligible influence on the quality of approximation according to (12). This value has to be determined separately for each particular

problem. The coefficients c_v , resulting from (15) by the use of this value, λ_{opt} , define the solution in real space according to (9), whereas the approximation curve to the data points is given by

$$\bar{I}_A(h) = \sum_{v=1}^N c_v \chi_v(h). \quad (17)$$

The desmeared scattering curve is defined by the equation

$$I_A(h) = \sum_{v=1}^N c_v \dot{\psi}_v(h). \quad (18)$$

The procedure for the solution of (7) can be summarized in the following way: (1) statement of an estimate for the largest distance D , (2) computation of spline functions $\varphi_v(r)$, (3) Fourier transformation of $\varphi_v(r)$; result: $\psi_v(h)$, (4) smearing of the Fourier-transformed splines $\psi_v(h)$, (5) computation of $B_{\mu\nu}$ and d_μ , (6) solution of (15) for a series of values of λ , (7) determination of λ_{opt} , (8) computation of the solution functions according to (9), (18) and (17) by use of the solution coefficients belonging to the solution of (15) for λ_{opt} .

A Fortran program* has been written for the execution of these points and for the calculation of some quantities correlated with $p_A(r)$.

* For further details of the program ITP and for program listings, please contact the author.

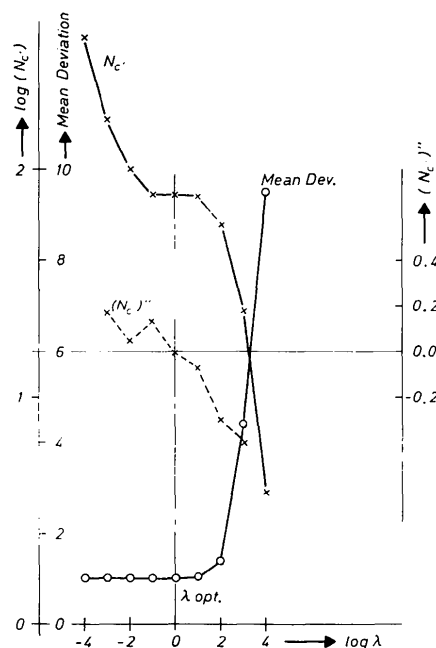


Fig. 2. Determination of the stabilization parameter, λ_{opt} , by the point of inflexion method. Dependency of $N_{c'}$ and mean deviation per point on λ .

Radius of gyration and zero intensity

The computation of the radius of gyration according to (2) with $p_A(r)$ gives the approximation

$$R_{gA}^2 = \int_0^D p_A(r) r^2 dr / 2 \int_0^D p_A(r) dr. \quad (19)$$

This approximation has the advantage that the radius of gyration is calculated from the smeared data points, with the whole scattering curve. For this reason, this approximation is often superior to the Guinier approximation. This point has been strengthened by a series of model calculations (see *Radius of gyration*).

An approximate value for the zero intensity can be computed according to (3):

$$I_A(0) = 4\pi \int_0^D p_A(r) dr. \quad (20)$$

Numerical results

The field of application, the limits and the accuracy of the method are demonstrated by means of some

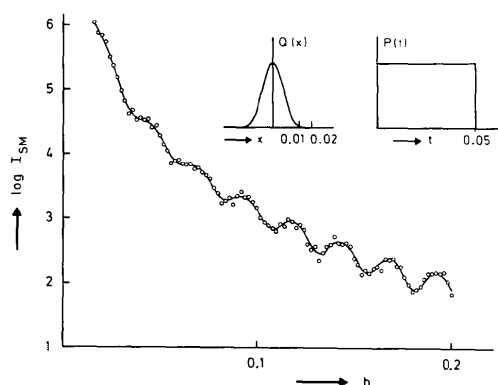


Fig. 3. Smeared scattering function of a sphere. — Theoretical smeared curve according to slit-width profile $Q(x)$ and slit-length profile $P(t)$. ···· Data points, 10% statistical error, $\Delta h R_g = 0.2$, $h_1 R_g = 1.6$, $h_2 R_g = 20$ and $R_g = 100$ Å.

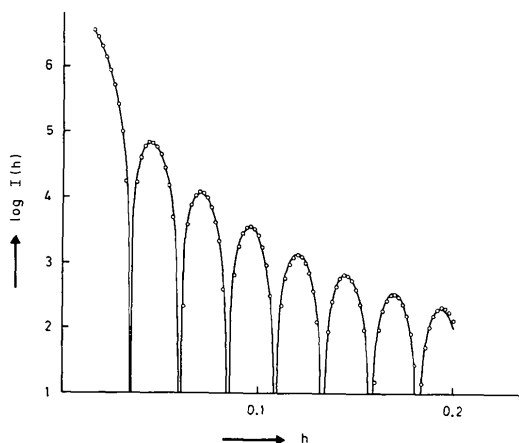


Fig. 4. Scattering function of a sphere, $R_g = 100$ Å. — Theoretical function, ○○○ desmeared data points from Fig. 3.

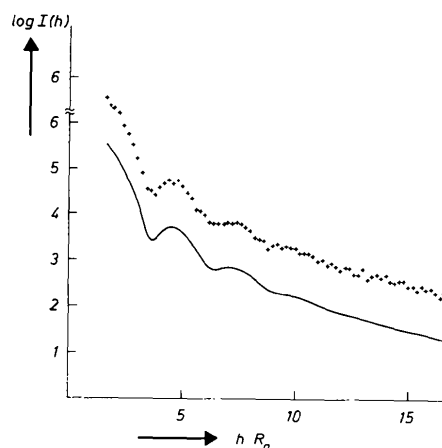


Fig. 5. Wavelength effect. Gaussian wavelength distribution with full half-width of about 24%. — Smeared scattering function of a sphere. + + + Data points with 10% statistical error, $\Delta h R_g = 0.2$, $h_1 R_g = 1.6$ and $h_2 R_g = 20$.

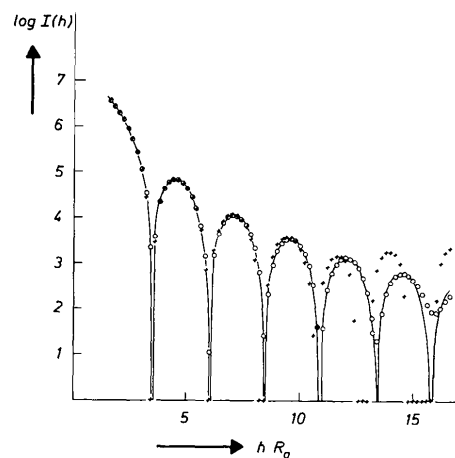


Fig. 6. Wavelength desmearing; — theoretical scattering function, ○○○ desmeared function resulting from the exact smeared function, + + + desmeared noisy data points.

selected examples. Detailed documentation is given elsewhere. A smeared scattering curve of a sphere is shown, together with the slit profiles $P(t)$ and $Q(x)$, in Fig. 3. The data points are generated by adding 10% statistical noise to the smeared curve. The desmeared curve from (18) is shown in Fig. 4. The termination effects appear to be small. The resolution of the maxima and minima is excellent. This resolution is of great importance to the subsequent calculation of Fourier transforms of the amplitudes. The exact diameter of the sphere is 258.2 Å ($R_g = 100$ Å); D is 300 Å. The Guinier approximation is not applicable, since the first data point is at $h_1 R_g = 1.6$. Application of (19) leads to an error of 1%. The deviations of $p_A(r)$ from $p(r)$ are small.

The number of neutron small-angle experiments is increasing rapidly. The next example shows an applica-

tion to the problem of scattering curves smeared by a wavelength distribution. The smeared curve (Gaussian wavelength distribution, with full half-width of about 24%) is plotted in Fig. 5, together with points generated by statistical fluctuations corresponding to 10% error. The desmeared curves and $p_A(r)$, together with the propagated error, are shown in Figs. 6 and 7, respectively. A comparison of the two curves in Fig. 6 shows that the calculated curve for data with statistical noise starts to diverge from the exact function in the region where the statistical oscillations are essentially larger than the oscillations of the exact smeared curve. It is absolutely necessary to use monochromators with a small half-width if neutron small-angle data are to be evaluated at relatively large angles (Engelman & Moore, 1975).

Several authors have mentioned the problems arising in data evaluation of scattering curves possessing a minimum at zero angle (Schmidt, 1970; Raynal, Schelten & Schmatz, 1971; Walter, Kranold & Becherer, 1974). The results for a Guinier-Walker model curve, as regarded by Walter *et al.* (1974), with $k=0.6$

$$\begin{aligned} \varrho(r) &= \varrho_2 & \text{for } 0 \leq r \leq kR & \quad \varrho_2/\varrho_1 = -4.0 \\ \varrho(r) &= \varrho_1 & \text{for } kR \leq r \leq R \\ \varrho(r) &= 0 & \text{for } r > R \end{aligned}$$

$\Delta(hR) = 0.125$ and 3% statistical error, are shown in Fig. 8.

Radius of gyration

The following numerical tests have been carried out: calculation of the radius of gyration for various types of bodies (sphere, cube, stick 1:1:10, plate 10:10:1 and chain molecules represented by the Debye formula) for point collimation, with and without statistical error, with variation of the beginning of the measuring range (h_1R_g) and comparison with the Guinier approximation.

The dependence of the accuracy of the results on h_1R_g and on the value of D is of special interest. Summarizing the results, it can be said that the error increases for both techniques [Guinier approximation and (19)] with increasing value of h_1R_g . The increment is the largest for elongated particles (chain molecule, stick). The absolute value of the error is always larger for the Guinier approximation.

The scattering curve of a chain molecule, approximated by the Debye formula

$$I(h) = \text{const.} \frac{2}{x^2} [e^{-x} - 1 + x] \quad (21)$$

(with $x = R_g^2 h^2$) with 5% statistical error, is demonstrated in Fig. 9 together with the Guinier plot. In this example, h_1R_g is chosen as 1.6. The ratio $I(h_1)/I(0)$ is about 0.5. The error resulting from the Guinier plot is larger than 25%. For $D = 4R_g$, the new method gives a radius of gyration with 2% error!

The dependence of the results of the new method on the choice of D is demonstrated with the scattering

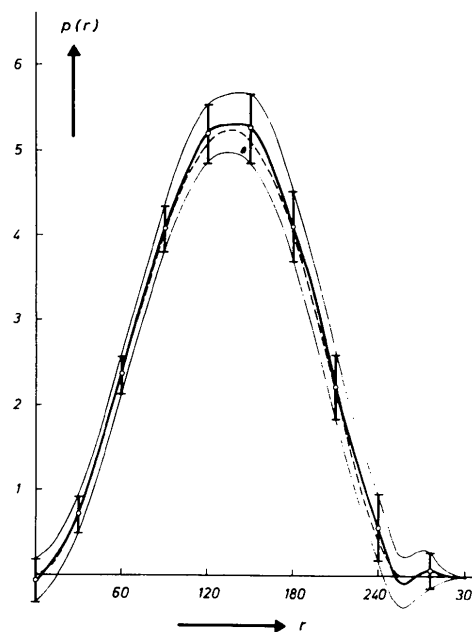


Fig. 7. $p(r)$ function of a sphere: ---- theoretical function, $\circ\circ\circ$ (with error bars) $p_A(r)$ calculated from the noisy data points (+ + +) in Fig. 5, together with the propagated error band.

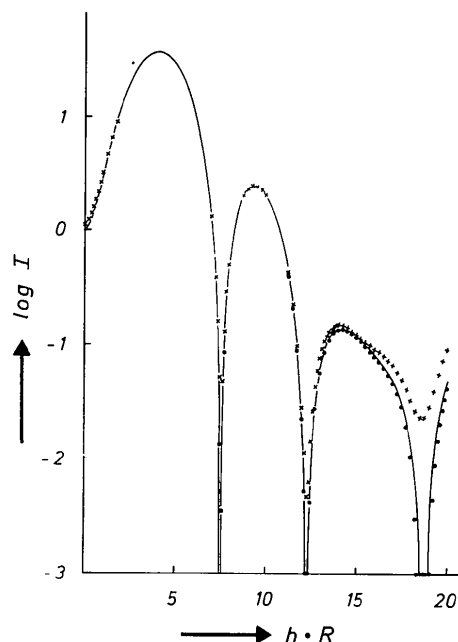


Fig. 8. Scattering function of a Guinier-Walker model ——. Data points with $\Delta hR = 0.125$, $h_1R = 0$ and $h_2R = 20$ generated by smearing according to an infinite long slit and desmeared with the program ITP. $\bullet\bullet\bullet$ From data without statistical noise, $\Delta R_{gA} = 0.3\%$, + + + from data with 3% statistical error, $\Delta R_{gA} = 0.5\%$.

curve of a stick. It should be noted that the results for more compact particles are always better (Müller, Laggner, Glatter & Kostner, 1977). The scattering data started with the large value $h_1 R_g = 1.6$, the spacing of the data points is $h R_g = 0.1$, the last point is located at $h_2 R_g = 10.0$ and the statistical noise has a magnitude of 5%. The distance D is varied from $2R_g$ up to $5R_g$. A slight systematic dependence of R_g results from the choice of D . The error is less than 10% over a relatively large region around $D_{\text{exact}} = 3.4R_g$ and is essentially smaller there than the error of the Guinier approximation (32%) (Fig. 10). The tests suggest that the choice of D is of importance for large $h_1 R_g$ values.

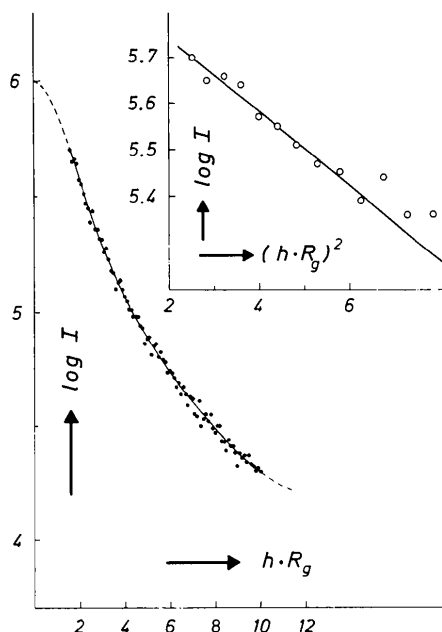


Fig. 9. Chain molecule, Debye formula: ----- theoretical function (point collimation). ●●● data points, 5% statistical error, $\Delta R_{gA} = 2\%$. Guinier plot: ○○○ data points, — Guinier approximation, $\Delta R_g > 25\%$.

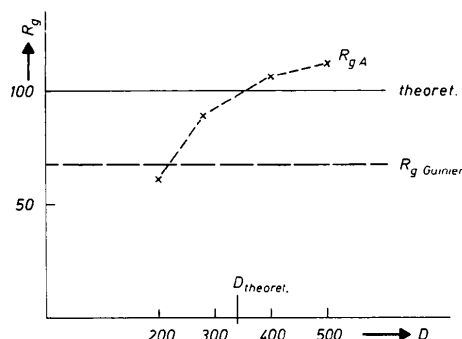


Fig. 10. Dependency of R_{gA} on the choice of D for a stick with $h_1 R_g = 1.6$, $\Delta h R_g = 0.1$ and $h_2 R_g = 10$.

Range of application

The distance distribution is always computed as a final result. This function is of great importance for the interpretation of the measurements. The corresponding error band represents the propagated error of the input data. If wavelength and collimation effects are not negligible, a smooth desmeared curve is computed as well. If systematic errors exist in the experimental data (scattering intensity, error band), or, if the assumptions, such as choice of D and number of spline functions N , are not consistent with the experimental data, there usually will be a breakdown of the method of inflexion point for the estimation of λ_{opt} . In such cases, the program gives an important hint about the inconsistency of the experimental data under the assumptions which are made.

Scattering curves with an additional background, when transformed by conventional Fourier transformation, may have strong oscillations in the $p(r)$ function. These oscillations can be of such magnitude that it is impossible to obtain any significant information from $p(r)$. Such oscillations do not occur with this new method. The background only causes a slight increase of $p(r)$ near r equal to zero.

A systematic limitation for the range of applicability can be caused by the type of function system used in (9) for the representation of $p(r)$. The number of functions N is limited for practical calculations to $N = 30$ (by the limited range of registration, by loss of information, by smearing effects and by statistical errors). For this reason, it may be possible that the fine structure of the function $p(r)$ cannot be resolved with adequate accuracy.

The method can consequently be used for (a) smoothing of scattering data, (b) desmearing, (c) weighted Fourier transformation with minimum termination effect, (d) estimation of radius of gyration and zero intensity, (e) control of quality and consistency of experimental data.

In principle, the applicability of this method is not restricted to monodisperse particle systems. Its application to polydisperse systems is equally possible. However, in its present version, the method does not give direct results concerning the particle-size distribution. The basic idea underlying this method allows an evaluation of the particle-size distribution from the original scattering data if a uniform particle shape can be assumed. With this aim, a modified version is being developed.

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